

Taming High-Dimensionality by Deep Learning

- ❑ Deep learning methods (e.g., autoencoder, feed-forward neural networks, CNNs, RNNs, GNNs) produce vector representation (i.e., embedding) of the input data tuples with a much smaller number of dimensions than the original ones, and thus naturally alleviate the high-dimensionality challenge
- ❑ Owing to deep learning's strong ability in learning semantically meaningful representation, the produced embedding often captures the complicate (e.g., non-linear) interaction between different input features, and thus is capable of detecting outliers that might be overlooked by the alternative methods (e.g., linear matrix factorization based approaches)

Using Deep Learning as Pre-processing

- ❑ Feed the input data set into an autoencoder, the output from the encoder produces the embedding of the input data tuples in a much lower dimensional space
- ❑ With such embedding as the input, we can use outlier detection methods (e.g., proximity-based approaches, reconstruction-based approaches) to detect outlying tuples
- ❑ Advantage: simple and flexible
- ❑ Disadvantage: separating the embedding learning stage and outlier detection stage might lead to sub-optimal detection results

Integrating Embedding Learning and Outlier Detection

- ❑ Find embedding of the input data tuples that is tailored for spotting the outliers, and simultaneously produce the embedding and detection results
 - ❑ If succeed, it often leads to a higher detection accuracy
- ❑ Example: one-class Neural Network (OC-NN)
 - ❑ Replace the hyperplane output (i.e., $w^T \phi(x)$) by a feed-forward neural network, whose output layer produces an outlier-ness score, and the last hidden layer produces the embedding of the input data
- ❑ Example: Deviation Networks (DevNet)
 - ❑ Replace the raw feature value x by its outlier-ness score $f(x, \Theta)$
 - ❑ $f(x, \Theta)$ is produced by a deep learning model (e.g., a feed-forward neural network with a linear output layer) with x as the input and Θ being the model parameters of the deep learning model
 - ❑ Replace μ and σ in z-normalization by the sample mean and the sample standard deviation of the outlier-ness scores of the normal tuples, respectively